Reg. No										

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MT509 – ABSTRACT ALGEBRA

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. Define subgroup.
2. Define normal subgroup.
3. Define even permutation.
4. State Pigeonhole principle.
5. Define Euclidean ring.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Let G be a group and H be a subgroup of G, then prove that the relation a $\equiv$ b mod H is an equivalence relation.
7. If H and K are subgroups of G and o(H) > $\sqrt{o(G)}$ and o(K) > $\sqrt{o(G)}$, then prove tha $H \cap K \neq \phi$.
8. Let G be a group of order 99, then show that H is a normal subgroup of order 11.
9. If R is a ring, then prove that a 0 = 0 a = 0 for all a in R.

10. Let R be the ring of all real valued continuous functions on the closed unit interval. Let M = {f(x) in R / f(1/2) = 0}, then show that M is a maximal ideal in R.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) i) Prove that there is a one to one correspondence between any two right cosets of H in G.
ii) If G is a finite group whose order is a prime p, then prove that G is a cyclic group.

(or)

b) If G is a group, then prove that
i) For every a in G, $(a^{-1})^{-1} = a$,
ii) For all a, b in G, $(a.b)^{-1} = b^{-1}.a^{-1}$.

12. a) Let ϕ be a homomorphism of G into $\bar{G}$ then prove that
i) $\phi(e) = \bar{e}$, the unit element of $\bar{G}$.
ii) $\phi(x^{-1}) = \phi(x)^{-1}$ for all x in G

(or)

b) Let ϕ be a homomorphism of G onto $\bar{G}$ with kernel K, then prove that $G / K \cong \bar{G}$.

13. a) If G is a group, then prove that the set of all automorphisms of G form a group.

(or)

b) If G is a group, H a subgroup of G, and S is the set of all right cosets of H in G, then prove that there is a homomorphism θ of G into A(S) and the kernel of θ is the largest normal subgroup of G which is contained in H.

14. a) If U is an ideal of the ring R, then show that R/U is a ring and is a homomorphic image of R.

(or)

b) If ϕ is a homomorphism of R into R' with kernel I(ϕ),
i) then show that $I(\phi)$ is a subgroup of R under addition.

ii) If a in $I(\phi)$ and r in R then show that both ar and ra are in $I(\phi)$.

15. a) If R is a commutative ring with unit element and M is and ideal of R, then show that M is maximal ideal of R if and only if R/M is a field.

(or)

b) State and prove unique factorization theorem.

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